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Bihar Engineering University, Patna

B.Tech. 3rd Semester Examination, 2023

Time: 03 Hours

Code: 101308 ***Subject: Surveying and Geomatics*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer of the following (any seven questions only): **[2x7=14]**

(a) The Bowditch method of adjusting a traverse is based on the assumptions that
- (i) Errors in linear measurements is proportional to its length
- (ii) Errors in angular measurements is inversely proportional to its length
- (iii) Both of above
- (iv) None of above

(b) The temporary adjustments of a prismatic compass are
1. Centering
2. Levelling
3. Focusing the prism

The correct order is
- (i) 1, 3 and 2
- (ii) 1, 2 and 3
- (iii) 2, 3 and 1
- (iv) 2, 1 and 3

(c) The correction for sag is
- (i) Always additive
- (ii) Always subtractive
- (iii) Always zero
- (iv) Sometimes additive and sometimes subtractive

(d) Local attraction in compass surveying may exit due to
- (i) Incorrect levelling of the magnetic needle
- (ii) Loss of magnetism of the needle
- (iii) Friction of the needle at the pivot
- (iv) Presence of magnetic substance near the instruments.

(e) If the quadrantal bearing of line is N25°W then whole circle bearing of the line is
- (i) 355°
- (ii) 205°
- (iii) 335°
- (iv) 295°

(f) In a well-conditioned triangle, no angle should be less than 30° and greater than 120°
- (i) True
- (ii) False

(g) In closed traverse surveying the sum of deflection angle will not be equal to 360°
- (i) True
- (ii) False

(h) In ___ of remote sensing, the Sun is primary source of energy.
- (i) Active
- (ii) Passive
- (iii) Both A and B
- (iv) Cannot say

(i) Which of the following is/are elements of visual interpretation?
- (i) Contrast stretch
- (ii) Texture
- (iii) Density slicing
- (iv) None of the above

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(j) The lines joining points of equal dip are called

(i) Aclinic lines
(ii) Isogonic lines
(iii) Agonic lines
(iv) Isoclinic lines

Q.2 Solve both questions : **[14]**

(a) Determine the number of photographs required to cover an area 25km×25km if the scale is 1 in 10,000 and the format is 230×230mm. Take longitudinal lap as 60% and the side lap as 30%.

(b) What is vertical aerial photograph? With respect to a vertical aerial photograph, discuss the following using a suitable diagram:
(i) Principal Point
(ii) Nadir Point
(iii) Iso-center
(iv) Camera Axis
(v) Principal Line

Q.3 Solve both questions : **[14]**

(a) Explain the difference between
(i) Chain surveying and Compass surveying
(ii) Whole circle bearing system and Reducing bearing system
(iii) Vernier Theodolite and Precise Optical Theodolite.

(b) Draw a neat sketch of a vernier theodolite. With the help of the sketch, explain the various parts of vernier theodolite.

Q.4 Solve both questions : **[14]**

(a) Two straight lines intersect at chainage 1150.50, and the angle of intersection is 60˚. If the radius of the curve is 500 m, determine
(i) Tangent distance
(ii) Length of the curve
(iii) Chainages of point of curvature and tangency
(iv) Length of the long chord
(v) Degree of curve
(vi) Apex distance and the mid-ordinates

(b) Explain the resection after orientation by
(i) Two points problem
(ii) Three points problem

Q.5 Solve both questions : **[14]**

(a) Define the following terms:
(i) Traverse
(ii) True meridian
(iii) Declination
(iv) Dip

(b) The bearing observed in traversing with a compass at G.E.C. Nawada where local attraction was suspected are given below:
Line AB: Fore Bearing 134˚30', Back Bearing 314˚30'
Line BC: Fore Bearing 120˚00', Back Bearing 299˚30'
Line CD: Fore Bearing 320˚00', Back Bearing 185˚30'
Line DA: Fore Bearing 265˚00', Back Bearing 83˚30'
Find the corrected bearing of the lines. At what stations do you suspect local attraction?

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Q.6 Solve both questions : **[14]**

(a) Describe various applications of remote sensing in the field of civil engineering with suitable examples.

(b) What are the different methods of GPS surveying? Explain any one of them in details.

Q.7 Solve both questions : **[14]**

(a) What are the different types of waves used in EDM? What are the principles of EDM?

(b) Discuss in brief digital image processing process. What are the advantages of digital image processing?

Q.8 Solve both questions : **[14]**

(a) Briefly explain the different methods adopting in levelling. Explain with neat sketch.

(b) The staff reading taken on a staff held at distance of 80 m from the instrument with the bubble central was 1.455 m. When the bubble is moved 6 divisions out of the centre, the staff reading observed is 1.487 m. If the length of one division is 2 mm. Find the radius of curvature and sensitivity of the tube.

Q.9 Solve both questions : (incomplete question due to distorted question pdf) **[14]**

(a) A line was measured with a steel tape which was exactly 30 m at a temperature of 30˚C and a pull of 20 kg. The measured length was 1950 m. The temperature during measurement was 40˚C and the pull applied was 25 kg. Find the true length of the line if the area of cross-section of the tape is $2.5mm^2$. The coefficient of thermal expansion of material of the tape and modulus of elasticity of the materials are given as standard.

(b) Determine the length of the line given the true area conditions.

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